

PW1: Object-relational databases

Domain description: ARENA

Your domain is: the arena.

The general business process of the arena is as follows: the event organizers submit the requests. The arena manager confirms the events. The arena provides the customer with infrastructure for the event. If there is no available supplies or infrastructure available at the arena (e.g., light or sound equipment), the arena has a list of contracted vendors that may be used for obtaining the equipment or services.

The arena also provides ticket purchasing for various events. For events, there can be assigned different price levels of tickets. For sports events in particular, the ticket price is automatically calculated based on “the base” price and coefficient (e.g., if seat of type A costs 10 EUR (base price) and coefficient is 1.5, seat of type B will cost 15 but if seat of type A cost 20 EUR (base price) with the same coefficient, the seat of type B may cost 30). For other events, the prices may be all the same, or be split based on the sectors (e.g., for concerts). One ticket number can be used to purchase multiple tickets, and tickets must be personalized.

The event planner plans how many tickets will be sold to which events, as well as plans supplies, services and infrastructure necessary for each event, and estimates the budget based on the cost. The arena workers connect to the database to see the list of non-outsourced supplies and infrastructure that they need to build, and input estimated, and actual hours needed for building the necessary infrastructure. The arena workers also may add issues and necessary fixes for the infrastructure that the arena owns.

The customer may subscribe for the entire season on silver, gold or VIP level, for a fixed price. These levels will get a person into various events: silver pass in silver events, gold: in silver and gold events, VIP pass – in all events. The customer interacts with the database via arena’s application (including ticket purchases).

The event organizers must be able to fill out the form that saves their requests in the database.